3. Anexos

A. Desbloqueo de los registros y MVEC

Para desbloquearlos escribir en `SYSKEY` los valores:

```
SYSKEY=0xAA996655;
SYSKEY=0x556699AA;
```

Para volverlos a bloquear escribir cualquier valor en `SYSKEY`.
El bit MVEC del registro `INTCON es el 12`.

B. Datasheets

$$UxBRG = \frac{f_{PBCLK}}{4^{(2-BRGH)} \cdot Baudrate} - 1$$

Registro de transmisión: `UxTXREG`
Registro de recepción: `UxRXREG`

REGISTER 19-1: UxMODE: UARTx MODE REGISTER (CONTINUED)

bit 5 **ABAUD:** Auto-Baud Enable bit
`1` = Enable baud rate measurement on the next character – requires reception of Sync character (0x55); cleared by hardware upon completion
`0` = Baud rate measurement disabled or completed

bit 4 **RXINV:** Receive Polarity Inversion bit
`1` = UxRX Idle state is '`0`'
`0` = UxRX Idle state is '`1`'

bit 3 **BRGH:** High Baud Rate Enable bit
`1` = High-Speed mode – 4x baud clock enabled
`0` = Standard Speed mode – 16x baud clock enabled

bit 2-1 **PDSEL<1:0>:** Parity and Data Selection bits
`11` = 9-bit data, no parity
`10` = 8-bit data, odd parity
`01` = 8-bit data, even parity
`00` = 8-bit data, no parity

bit 0 **STSEL:** Stop Selection bit
`1` = 2 Stop bits
`0` = 1 Stop bit

Note 1: When using 1:1 PBCLK divisor, the user software should not read/write the peripheral SFRs in the SYSCLK cycle immediately following the instruction that clears the module's ON bit.

REGISTER 19-1: UxMODE: UARTx MODE REGISTER

Bit Range	Bit 31/23/15/7	Bit 30/22/14/6	Bit 29/21/13/5	Bit 28/20/12/4	Bit 27/19/11/3	Bit 26/18/10/2	Bit 25/17/9/1	Bit 24/16/8/0
31:24	U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
	—	—	—	—	—	—	—	—
23:16	U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
	—	—	—	—	—	—	—	—
15:8	R/W-0	U-0	R/W-0	R/W-0	R/W-0	U-0	R/W-0	R/W-0
	ON(1)	—	SIDL	IREN	RTSMD	—	UEN<1:0>	
7:0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
	WAKE	LPBACK	ABAUD	RXINV	BRGH	PDSEL<1:0>		STSEL

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

bit 31-16 **Unimplemented:** Read as '0'

bit 15 **ON:** UARTx Enable bit(1)
- 1 = UARTx is enabled. UARTx pins are controlled by UARTx as defined by the UEN<1:0> and UTXEN control bits.
- 0 = UARTx is disabled. All UARTx pins are controlled by corresponding bits in the PORTx, TRISx and LATx registers; UARTx power consumption is minimal.

bit 14 **Unimplemented:** Read as '0'

bit 13 **SIDL:** Stop in Idle Mode bit
- 1 = Discontinue module operation when the device enters Idle mode
- 0 = Continue module operation when the device enters Idle mode

bit 12 **IREN:** IrDA Encoder and Decoder Enable bit
- 1 = IrDA is enabled
- 0 = IrDA is disabled

bit 11 **RTSMD:** Mode Selection for $\overline{\text{UxRTS}}$ Pin bit
- 1 = $\overline{\text{UxRTS}}$ pin is in Simplex mode
- 0 = $\overline{\text{UxRTS}}$ pin is in Flow Control mode

bit 10 **Unimplemented:** Read as '0'

bit 9-8 **UEN<1:0>:** UARTx Enable bits
- 11 = UxTX, UxRX and UxBCLK pins are enabled and used; $\overline{\text{UxCTS}}$ pin is controlled by corresponding bits in the PORTx register
- 10 = UxTX, UxRX, $\overline{\text{UxCTS}}$ and $\overline{\text{UxRTS}}$ pins are enabled and used
- 01 = UxTX, UxRX and $\overline{\text{UxRTS}}$ pins are enabled and used; $\overline{\text{UxCTS}}$ pin is controlled by corresponding bits in the PORTx register
- 00 = UxTX and UxRX pins are enabled and used; $\overline{\text{UxCTS}}$ and $\overline{\text{UxRTS}}$/UxBCLK pins are controlled by corresponding bits in the PORTx register

bit 7 **WAKE:** Enable Wake-up on Start bit Detect During Sleep Mode bit
- 1 = Wake-up enabled
- 0 = Wake-up disabled

bit 6 **LPBACK:** UARTx Loopback Mode Select bit
- 1 = Loopback mode is enabled
- 0 = Loopback mode is disabled

Note 1: When using 1:1 PBCLK divisor, the user software should not read/write the peripheral SFRs in the SYSCLK cycle immediately following the instruction that clears the module's ON bit.

REGISTER 19-2: UxSTA: UARTx STATUS AND CONTROL REGISTER

Bit Range	Bit 31/23/15/7	Bit 30/22/14/6	Bit 29/21/13/5	Bit 28/20/12/4	Bit 27/19/11/3	Bit 26/18/10/2	Bit 25/17/9/1	Bit 24/16/8/0
31:24	U-0	U-0	U-0	U-0	U-0	U-0	U-0	R/W-0
	—	—	—	—	—	—	—	ADM_EN
23:16	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
	ADDR<7:0>							
15:8	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R-0	R-1
	UTXISEL<1:0>		UTXINV	URXEN	UTXBRK	UTXEN	UTXBF	TRMT
7:0	R/W-0	R/W-0	R/W-0	R-1	R-0	R-0	R/W-0	R-0
	URXISEL<1:0>		ADDEN	RIDLE	PERR	FERR	OERR	URXDA

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

bit 31-25 **Unimplemented:** Read as '0'

bit 24 **ADM_EN:** Automatic Address Detect Mode Enable bit
1 = Automatic Address Detect mode is enabled
0 = Automatic Address Detect mode is disabled

bit 23-16 **ADDR<7:0>:** Automatic Address Mask bits
When the ADM_EN bit is '1', this value defines the address character to use for automatic address detection.

bit 15-14 **UTXISEL<1:0>:** TX Interrupt Mode Selection bits
11 = Reserved, do not use
10 = Interrupt is generated and asserted while the transmit buffer is empty
01 = Interrupt is generated and asserted when all characters have been transmitted
00 = Interrupt is generated and asserted while the transmit buffer contains at least one empty space

bit 13 **UTXINV:** Transmit Polarity Inversion bit
If IrDA mode is disabled (i.e., IREN (UxMODE<12>) is '0'):
1 = UxTX Idle state is '0'
0 = UxTX Idle state is '1'

If IrDA mode is enabled (i.e., IREN (UxMODE<12>) is '1'):
1 = IrDA encoded UxTX Idle state is '1'
0 = IrDA encoded UxTX Idle state is '0'

bit 12 **URXEN:** Receiver Enable bit
1 = UARTx receiver is enabled. UxRX pin is controlled by UARTx (if ON = 1)
0 = UARTx receiver is disabled. UxRX pin is ignored by the UARTx module. UxRX pin is controlled by port.

bit 11 **UTXBRK:** Transmit Break bit
1 = Send Break on next transmission. Start bit followed by twelve '0' bits, followed by Stop bit; cleared by hardware upon completion
0 = Break transmission is disabled or completed

bit 10 **UTXEN:** Transmit Enable bit
1 = UARTx transmitter is enabled. UxTX pin is controlled by UARTx (if ON = 1).
0 = UARTx transmitter is disabled. Any pending transmission is aborted and buffer is reset. UxTX pin is controlled by port.

bit 9 **UTXBF:** Transmit Buffer Full Status bit (read-only)
1 = Transmit buffer is full
0 = Transmit buffer is not full, at least one more character can be written

bit 8 **TRMT:** Transmit Shift Register is Empty bit (read-only)
1 = Transmit shift register is empty and transmit buffer is empty (the last transmission has completed)
0 = Transmit shift register is not empty, a transmission is in progress or queued in the transmit buffer

REGISTER 19-2: UxSTA: UARTx STATUS AND CONTROL REGISTER (CONTINUED)

bit 7-6 **URXISEL<1:0>:** Receive Interrupt Mode Selection bit
11 = Reserved; do not use
10 = Interrupt flag bit is asserted while receive buffer is 3/4 or more full (i.e., has 6 or more data characters)
01 = Interrupt flag bit is asserted while receive buffer is 1/2 or more full (i.e., has 4 or more data characters)
00 = Interrupt flag bit is asserted while receive buffer is not empty (i.e., has at least 1 data character)

bit 5 **ADDEN:** Address Character Detect bit (bit 8 of received data = 1)
1 = Address Detect mode is enabled. If 9-bit mode is not selected, this control bit has no effect.
0 = Address Detect mode is disabled

bit 4 **RIDLE:** Receiver Idle bit (read-only)
1 = Receiver is Idle
0 = Data is being received

bit 3 **PERR:** Parity Error Status bit (read-only)
1 = Parity error has been detected for the current character
0 = Parity error has not been detected

bit 2 **FERR:** Framing Error Status bit (read-only)
1 = Framing error has been detected for the current character
0 = Framing error has not been detected

bit 1 **OERR:** Receive Buffer Overrun Error Status bit.
This bit is set in hardware and can only be cleared (= 0) in software. Clearing a previously set OERR bit resets the receiver buffer and the RSR to an empty state.
1 = Receive buffer has overflowed
0 = Receive buffer has not overflowed

bit 0 **URXDA:** Receive Buffer Data Available bit (read-only)
1 = Receive buffer has data, at least one more character can be read
0 = Receive buffer is empty

Interrupt Source(1)	IRQ #	Vector #	Interrupt Bit Location				Persistent Interrupt
			Flag	Enable	Priority	Sub-priority	
U1E – UART1 Fault	39	32	IFS1<7>	IEC1<7>	IPC8<4:2>	IPC8<1:0>	Yes
U1RX – UART1 Receive Done	40	32	IFS1<8>	IEC1<8>	IPC8<4:2>	IPC8<1:0>	Yes
U1TX – UART1 Transfer Done	41	32	IFS1<9>	IEC1<9>	IPC8<4:2>	IPC8<1:0>	Yes
I2C1B – I2C1 Bus Collision Event	42	33	IFS1<10>	IEC1<10>	IPC8<12:10>	IPC8<9:8>	Yes
U2E – UART2 Error	53	37	IFS1<21>	IEC1<21>	IPC9<12:10>	IPC9<9:8>	Yes
U2RX – UART2 Receiver	54	37	IFS1<22>	IEC1<22>	IPC9<12:10>	IPC9<9:8>	Yes
U2TX – UART2 Transmitter	55	37	IFS1<23>	IEC1<23>	IPC9<12:10>	IPC9<9:8>	Yes

Interrupt Source(1)	IRQ #	Vector #	Interrupt Bit Location				Persistent Interrupt
			Flag	Enable	Priority	Sub-priority	
Highest Natural Order Priority							
CT – Core Timer Interrupt	0	0	IFS0<0>	IEC0<0>	IPC0<4:2>	IPC0<1:0>	No
CS0 – Core Software Interrupt 0	1	1	IFS0<1>	IEC0<1>	IPC0<12:10>	IPC0<9:8>	No
CS1 – Core Software Interrupt 1	2	2	IFS0<2>	IEC0<2>	IPC0<20:18>	IPC0<17:16>	No
INT0 – External Interrupt	3	3	IFS0<3>	IEC0<3>	IPC0<28:26>	IPC0<25:24>	No
T1 – Timer1	4	4	IFS0<4>	IEC0<4>	IPC1<4:2>	IPC1<1:0>	No
IC1E – Input Capture 1 Error	5	5	IFS0<5>	IEC0<5>	IPC1<12:10>	IPC1<9:8>	Yes
IC1 – Input Capture 1	6	5	IFS0<6>	IEC0<6>	IPC1<12:10>	IPC1<9:8>	Yes
OC1 – Output Compare 1	7	6	IFS0<7>	IEC0<7>	IPC1<20:18>	IPC1<17:16>	No
INT1 – External Interrupt 1	8	7	IFS0<8>	IEC0<8>	IPC1<28:26>	IPC1<25:24>	No
T2 – Timer2	9	8	IFS0<9>	IEC0<9>	IPC2<4:2>	IPC2<1:0>	No
IC2E – Input Capture 2	10	9	IFS0<10>	IEC0<10>	IPC2<12:10>	IPC2<9:8>	Yes
IC2 – Input Capture 2	11	9	IFS0<11>	IEC0<11>	IPC2<12:10>	IPC2<9:8>	Yes
OC2 – Output Compare 2	12	10	IFS0<12>	IEC0<12>	IPC2<20:18>	IPC2<17:16>	No
INT2 – External Interrupt 2	13	11	IFS0<13>	IEC0<13>	IPC2<28:26>	IPC2<25:24>	No
T3 – Timer3	14	12	IFS0<14>	IEC0<14>	IPC3<4:2>	IPC3<1:0>	No
IC3E – Input Capture 3	15	13	IFS0<15>	IEC0<15>	IPC3<12:10>	IPC3<9:8>	Yes
IC3 – Input Capture 3	16	13	IFS0<16>	IEC0<16>	IPC3<12:10>	IPC3<9:8>	Yes
OC3 – Output Compare 3	17	14	IFS0<17>	IEC0<17>	IPC3<20:18>	IPC3<17:16>	No
INT3 – External Interrupt 3	18	15	IFS0<18>	IEC0<18>	IPC3<28:26>	IPC3<25:24>	No
T4 – Timer4	19	16	IFS0<19>	IEC0<19>	IPC4<4:2>	IPC4<1:0>	No
IC4E – Input Capture 4 Error	20	17	IFS0<20>	IEC0<20>	IPC4<12:10>	IPC4<9:8>	Yes
IC4 – Input Capture 4	21	17	IFS0<21>	IEC0<21>	IPC4<12:10>	IPC4<9:8>	Yes
OC4 – Output Compare 4	22	18	IFS0<22>	IEC0<22>	IPC4<20:18>	IPC4<17:16>	No
INT4 – External Interrupt 4	23	19	IFS0<23>	IEC0<23>	IPC4<28:26>	IPC4<25:24>	No
T5 – Timer5	24	20	IFS0<24>	IEC0<24>	IPC5<4:2>	IPC5<1:0>	No
IC5E – Input Capture 5 Error	25	21	IFS0<25>	IEC0<25>	IPC5<12:10>	IPC5<9:8>	Yes
IC5 – Input Capture 5	26	21	IFS0<26>	IEC0<26>	IPC5<12:10>	IPC5<9:8>	Yes
OC5 – Output Compare 5	27	22	IFS0<27>	IEC0<27>	IPC5<20:18>	IPC5<17:16>	No
AD1 – ADC1 Convert done	28	23	IFS0<28>	IEC0<28>	IPC5<28:26>	IPC5<25:24>	Yes
FSCM – Fail-Safe Clock Monitor	29	24	IFS0<29>	IEC0<29>	IPC6<4:2>	IPC6<1:0>	No
RTCC – Real-Time Clock and Calendar	30	25	IFS0<30>	IEC0<30>	IPC6<12:10>	IPC6<9:8>	No
FCE – Flash Control Event	31	26	IFS0<31>	IEC0<31>	IPC6<20:18>	IPC6<17:16>	No
CMP1 – Comparator Interrupt	32	27	IFS1<0>	IEC1<0>	IPC6<28:26>	IPC6<25:24>	No
CMP2 – Comparator Interrupt	33	28	IFS1<1>	IEC1<1>	IPC7<4:2>	IPC7<1:0>	No
CMP3 – Comparator Interrupt	34	29	IFS1<2>	IEC1<2>	IPC7<12:10>	IPC7<9:8>	No
USB – USB Interrupts	35	30	IFS1<3>	IEC1<3>	IPC7<20:18>	IPC7<17:16>	Yes
SPI1E – SPI1 Fault	36	31	IFS1<4>	IEC1<4>	IPC7<28:26>	IPC7<25:24>	Yes
SPI1RX – SPI1 Receive Done	37	31	IFS1<5>	IEC1<5>	IPC7<28:26>	IPC7<25:24>	Yes
SPI1TX – SPI1 Transfer Done	38	31	IFS1<6>	IEC1<6>	IPC7<28:26>	IPC7<25:24>	Yes

REGISTER 13-1: TXCON: TYPE B TIMER CONTROL REGISTER

Bit Range	Bit 31/23/15/7	Bit 30/22/14/6	Bit 29/21/13/5	Bit 28/20/12/4	Bit 27/19/11/3	Bit 26/18/10/2	Bit 25/17/9/1	Bit 24/16/8/0
31:24	U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
	—	—	—	—	—	—	—	—
23:16	U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
	—	—	—	—	—	—	—	—
15:8	R/W-0	U-0	R/W-0	U-0	U-0	U-0	U-0	U-0
	ON(1,3)	—	SIDL(4)	—	—	—	—	—
7:0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	U-0	R/W-0	U-0
	TGATE(3)	TCKPS<2:0>(3)			T32(2)	—	TCS(3)	—

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as ‘0’	
-n = Value at POR	‘1’ = Bit is set	‘0’ = Bit is cleared	x = Bit is unknown

bit 31-16 **Unimplemented:** Read as ‘0’

bit 15 **ON:** Timer On bit(1,3)
1 = Module is enabled
0 = Module is disabled

bit 14 **Unimplemented:** Read as ‘0’

bit 13 **SIDL:** Stop in Idle Mode bit(4)
1 = Discontinue module operation when the device enters Idle mode
0 = Continue module operation when the device enters Idle mode

bit 12-8 **Unimplemented:** Read as ‘0’

bit 7 **TGATE:** Timer Gated Time Accumulation Enable bit(3)
When TCS = 1:
This bit is ignored and is read as ‘0’.
When TCS = 0:
1 = Gated time accumulation is enabled
0 = Gated time accumulation is disabled

bit 6-4 **TCKPS<2:0>:** Timer Input Clock Prescale Select bits(3)
111 = 1:256 prescale value
110 = 1:64 prescale value
101 = 1:32 prescale value
100 = 1:16 prescale value
011 = 1:8 prescale value
010 = 1:4 prescale value
001 = 1:2 prescale value
000 = 1:1 prescale value

REGISTER 13-1: TXCON: TYPE B TIMER CONTROL REGISTER (CONTINUED)

bit 3 **T32:** 32-Bit Timer Mode Select bit(2)
1 = Odd numbered and even numbered timers form a 32-bit timer
0 = Odd numbered and even numbered timers form a separate 16-bit timer

bit 2 **Unimplemented:** Read as ‘0’

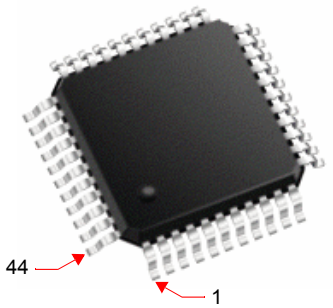
bit 1 **TCS:** Timer Clock Source Select bit(3)
1 = External clock from TxCK pin
0 = Internal peripheral clock

bit 0 **Unimplemented:** Read as ‘0’

TABLE 12: PIN NAMES FOR 44-PIN USB DEVICES

44-PIN TQFP (TOP VIEW)(1,2,3,5)

PIC32MX210F016D
PIC32MX220F032D
PIC32MX230F064D
PIC32MX230F256D
PIC32MX250F128D
PIC32MX270F256D



Pin #	Full Pin Name	Pin #	Full Pin Name
1	RPB9/SDA1/CTED4/PMD3/RB9	23	AN4/C1INB/C2IND/RPB2/SDA2/CTED13/PMD2/RB2
2	RPC6/PMA1/RC6	24	AN5/C1INA/C2INC/RTCC/RPB3/SCL2/PMWR/RB3
3	RPC7/PMA0/RC7	25	AN6/RPC0/RC0
4	RPC8/PMA5/RC8	26	AN7/RPC1/RC1
5	RPC9/CTED7/PMA6/RC9	27	AN8/RPC2/PMA2/RC2
6	V_{SS}	28	V_{DD}
7	V_{CAP}	29	V_{SS}
8	PGED2/RPB10/D+/CTED11/RB10	30	OSC1/CLKI/RPA2/RA2
9	PGEC2/RPB11/D-/RB11	31	OSC2/CLKO/RPA3/RA3
10	V_{USB3V3}	32	TDO/RPA8/PMA8/RA8
11	AN11/RPB13/CTPLS/PMRD/RB13	33	SOSCI/RPB4/RB4
12	PGED4(4)/TMS/PMA10/RA10	34	SOSCO/RPA4/T1CK/CTED9/RA4
13	PGEC4(4)/TCK/CTED8/PMA7/RA7	35	TDI/RPA9/PMA9/RA9
14	CV_{REFOUT}/AN10/C3INB/RPB14/V_{BUSON}/SCK1/CTED5/RB14	36	AN12/RPC3/RC3
15	AN9/C3INA/RPB15/SCK2/CTED6/PMCS1/RB15	37	RPC4/PMA4/RC4
16	AV_{SS}	38	RPC5/PMA3/RC5
17	AV_{DD}	39	V_{SS}
18	$\overline{MCLR}$	40	V_{DD}
19	PGED3/V_{REF}+/CV_{REF}+/AN0/C3INC/RPA0/CTED1/PMD7/RA0	41	RPB5/USBID/RB5
20	PGEC3/V_{REF}-/CV_{REF}-/AN1/RPA1/CTED2/PMD6/RA1	42	V_{BUS}
21	PGED1/AN2/C1IND/C2INB/C3IND/RPB0/PMD0/RB0	43	RPB7/CTED3/PMD5/INT0/RB7
22	PGEC1/AN3/C1INC/C2INA/RPB1/CTED12/PMD1/RB1	44	RPB8/SCL1/CTED10/PMD4/RB8

Note 1: The RPn pins can be used by remappable peripherals. See Table 1 for the available peripherals and **Section 11.3 "Peripheral Pin Select"** for restrictions.
2: Every I/O port pin (RAx-RCx) can be used as a change notification pin (CNAx-CNCx). See **Section 11.0 "I/O Ports"** for more information.
3: The metal plane at the bottom of the device is not connected to any pins and is recommended to be connected to V_{SS} externally.
4: This pin function is not available on PIC32MX210F016D and PIC32MX220F032D devices.
5: Shaded pins are 5V tolerant.

TABLE 11-1: INPUT PIN SELECTION

Peripheral Pin	[*pin name*]R SFR	[*pin name*]R bits	[*pin name*]R Value to RPn Pin Selection
INT4	INT4R	INT4R<3:0>	0000 = RPA0 0001 = RPB3 0010 = RPB4 0011 = RPB15 0100 = RPB7 0101 = RPC7(2) 0110 = RPC0(1) 0111 = RPC5(2) 1000 = Reserved • • • 1111 = Reserved
T2CK	T2CKR	T2CKR<3:0>	
IC4	IC4R	IC4R<3:0>	
$\overline{\text{SS1}}$	SS1R	SS1R<3:0>	
REFCLKI	REFCLKIR	REFCLKIR<3:0>	
INT3	INT3R	INT3R<3:0>	0000 = RPA1 0001 = RPB5 0010 = RPB1 0011 = RPB11 0100 = RPB8 0101 = RPA8(2) 0110 = RPC8(2) 0111 = RPA9(2) 1000 = Reserved • • • 1111 = Reserved
T3CK	T3CKR	T3CKR<3:0>	
IC3	IC3R	IC3R<3:0>	
$\overline{\text{U1CTS}}$	U1CTSR	U1CTSR<3:0>	
U2RX	U2RXR	U2RXR<3:0>	
SDI1	SDI1R	SDI1R<3:0>	
INT2	INT2R	INT2R<3:0>	0000 = RPA2 0001 = RPB6 0010 = RPA4 0011 = RPB13 0100 = RPB2 0101 = RPC6(2) 0110 = RPC1(1) 0111 = RPC3(1) 1000 = Reserved • • • 1111 = Reserved
T4CK	T4CKR	T4CKR<3:0>	
IC1	IC1R	IC1R<3:0>	
IC5	IC5R	IC5R<3:0>	
U1RX	U1RXR	U1RXR<3:0>	
$\overline{\text{U2CTS}}$	U2CTSR	U2CTSR<3:0>	
SDI2	SDI2R	SDI2R<3:0>	
OCFB	OCFBR	OCFBR<3:0>	
INT1	INT1R	INT1R<3:0>	0000 = RPA3 0001 = RPB14 0010 = RPB0 0011 = RPB10 0100 = RPB9 0101 = RPC9(1) 0110 = RPC2(2) 0111 = RPC4(2) 1000 = Reserved • • • 1111 = Reserved
T5CK	T5CKR	T5CKR<3:0>	
IC2	IC2R	IC2R<3:0>	
$\overline{\text{SS2}}$	SS2R	SS2R<3:0>	
OCFA	OCFAR	OCFAR<3:0>	

Note 1: This pin is not available on 28-pin devices.
2: This pin is only available on 44-pin devices.

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TABLE 11-2: OUTPUT PIN SELECTION

RPn Port Pin	RPnR SFR	RPnR bits	RPnR Value to Peripheral Selection
RPA0	RPA0R	RPA0R<3:0>	0000 = No Connect 0001 = U1TX 0010 = U2RTS 0011 = SS1 0100 = Reserved 0101 = OC1 0110 = Reserved 0111 = C2OUT 1000 = Reserved . . . 1111 = Reserved
RPB3	RPB3R	RPB3R<3:0>	
RPB4	RPB4R	RPB4R<3:0>	
RPB15	RPB15R	RPB15R<3:0>	
RPB7	RPB7R	RPB7R<3:0>	
RPC7	RPC7R	RPC7R<3:0>	
RPC0	RPC0R	RPC0R<3:0>	
RPC5	RPC5R	RPC5R<3:0>	
RPA1	RPA1R	RPA1R<3:0>	0000 = No Connect 0001 = Reserved 0010 = Reserved 0011 = SDO1 0100 = SDO2 0101 = OC2 0110 = Reserved 0111 = C3OUT . . . 1111 = Reserved
RPB5	RPB5R	RPB5R<3:0>	
RPB1	RPB1R	RPB1R<3:0>	
RPB11	RPB11R	RPB11R<3:0>	
RPB8	RPB8R	RPB8R<3:0>	
RPA8	RPA8R	RPA8R<3:0>	
RPC8	RPC8R	RPC8R<3:0>	
RPA9	RPA9R	RPA9R<3:0>	
RPA2	RPA2R	RPA2R<3:0>	0000 = No Connect 0001 = Reserved 0010 = Reserved 0011 = SDO1 0100 = SDO2 0101 = OC4 0110 = OC5 0111 = REFCLKO 1000 = Reserved . . . 1111 = Reserved
RPB6	RPB6R	RPB6R<3:0>	
RPA4	RPA4R	RPA4R<3:0>	
RPB13	RPB13R	RPB13R<3:0>	
RPB2	RPB2R	RPB2R<3:0>	
RPC6	RPC6R	RPC6R<3:0>	
RPC1	RPC1R	RPC1R<3:0>	
RPC3	RPC3R	RPC3R<3:0>	
RPA3	RPA3R	RPA3R<3:0>	0000 = No Connect 0001 = U1RTS 0010 = U2TX 0011 = Reserved 0100 = SS2 0101 = OC3 0110 = Reserved 0111 = C1OUT 1000 = Reserved . . . 1111 = Reserved
RPB14	RPB14R	RPB14R<3:0>	
RPB0	RPB0R	RPB0R<3:0>	
RPB10	RPB10R	RPB10R<3:0>	
RPB9	RPB9R	RPB9R<3:0>	
RPC9	RPC9R	RPC9R<3:0>	
RPC2	RPC2R	RPC2R<3:0>	
RPC4	RPC4R	RPC4R<3:0>	

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REGISTER 16-1: OCxCON: OUTPUT COMPARE 'x' CONTROL REGISTER

Bit Range	Bit 31/23/15/7	Bit 30/22/14/6	Bit 29/21/13/5	Bit 28/20/12/4	Bit 27/19/11/3	Bit 26/18/10/2	Bit 25/17/9/1	Bit 24/16/8/0
31:24	U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
	—	—	—	—	—	—	—	—
23:16	U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
	—	—	—	—	—	—	—	—
15:8	R/W-0	U-0	R/W-0	U-0	U-0	U-0	U-0	U-0
	ON(1)	—	SIDL	—	—	—	—	—
7:0	U-0	U-0	R/W-0	R-0	R/W-0	R/W-0	R/W-0	R/W-0
	—	—	OC32	OCFLT(2)	OCTSEL	OCM<2:0>		

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

bit 31-16 **Unimplemented:** Read as '0'

bit 15 **ON:** Output Compare Peripheral On bit(1)
1 = Output Compare peripheral is enabled
0 = Output Compare peripheral is disabled

bit 14 **Unimplemented:** Read as '0'

bit 13 **SIDL:** Stop in Idle Mode bit
1 = Discontinue module operation when the device enters Idle mode
0 = Continue module operation when the device enters Idle mode

bit 12-6 **Unimplemented:** Read as '0'

bit 5 **OC32:** 32-bit Compare Mode bit
1 = OCxR<31:0> and/or OCxRS<31:0> are used for comparisons to the 32-bit timer source
0 = OCxR<15:0> and OCxRS<15:0> are used for comparisons to the 16-bit timer source

bit 4 **OCFLT:** PWM Fault Condition Status bit(2)
1 = PWM Fault condition has occurred (cleared in hardware only)
0 = No PWM Fault condition has occurred

bit 3 **OCTSEL:** Output Compare Timer Select bit
1 = Timer3 is the clock source for this Output Compare module
0 = Timer2 is the clock source for this Output Compare module

bit 2-0 **OCM<2:0>:** Output Compare Mode Select bits
111 = PWM mode on OCx; Fault pin enabled
110 = PWM mode on OCx; Fault pin disabled
101 = Initialize OCx pin low; generate continuous output pulses on OCx pin
100 = Initialize OCx pin low; generate single output pulse on OCx pin
011 = Compare event toggles OCx pin
010 = Initialize OCx pin high; compare event forces OCx pin low
001 = Initialize OCx pin low; compare event forces OCx pin high
000 = Output compare peripheral is disabled but continues to draw current

Note 1: When using 1:1 PBCLK divisor, the user's software should not read/write the peripheral's SFRs in the SYSCLK cycle immediately following the instruction that clears the module's ON bit.

2: This bit is only used when OCM<2:0> = '111'. It is read as '0' in all other modes.